Create Azure Local VMs Enabled by Azure Arc

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Agenda

- **Objective:** Review deploying Azure Local VMs Enabled by Azure Arc
- **Scope:** Architecture, VM Types, Deployment Flow, and Management Capabilities

VM Types on Azure Local

- Azure Local supports three types of VMs:
  - Azure Local VMs enabled by Azure Arc
  - Azure Arc-enabled Servers
  - Unmanaged VMs

VM Type Comparison

- Azure Local VMs enabled by Azure Arc
  - Created via **Azure portal**, **Azure CLI**, or **ARM/Bicep**
  - Fully projected into Azure via the **Azure Arc resource bridge**
  - Support lifecycle operations: **start**, **stop**, **resize**, **disk attach**, **NIC attach**, etc.
- Azure Arc-enabled Servers
  - Physical or virtual machines connected to Azure using the **Connected Machine Agent**
  - No platform-level VM lifecycle controls
  - Support management & governance extensions only
- Unmanaged VMs
  - Created solely with **on-prem management tools**
  - Not synchronized with Azure
  - No Azure-based management

VM Provisioning Comparison

- **Arc-enabled VMs:** Provisioned in Azure; deployed locally
- **Arc-enabled servers:** Created via Hyper-V/Failover Cluster Manager/System Center and connected to Azure
- **Unmanaged VMs:** Created via Hyper-V/Failover Cluster Manager/System Center

VM Management Capability Comparison

- Arc-enabled Azure Local VMs support:
  - Start / Stop / Restart / Pause
  - Configuration changes (CPU, Memory)
  - Disk operations (add/remove/expand)
  - NIC operations (add/remove)
  - GPU attach/detach
  - Azure Monitoring, Defender for Cloud, Policies, Extensions
- Other VM types support partial or no lifecycle management.

OS Version Support

- Windows OS (portal & guest management)
  - Windows Server 2016, 2019, 2022, and 2025
  - Windows 10 and 11 Enterprise editions
- Windows Server 2012 / 2012 R2
  - Deployment allowed exclusively via Azure CLI using additional parameters
  - Guest management cannot be enabled due to lack of Hyper-V sockets support
- Windows Server 2008 R2 Status
  - Entirely unsupported for native Azure Local VM provisioning
- Linux OS (portal & guest management)
  - Ubuntu Server 20.04 LTS and 22.04 LTS
  - Red Hat Enterprise Linux (RHEL) 8 and 9
  - Rocky Linux 8 and 9
  - AlmaLinux 8 and 9
  - Mariner / Azure Linux

Core Component: Azure Arc Resource Bridge

- Prepackaged VM deployed automatically during Azure Local installation
- Projects local resources into Azure
- Acts as the management cluster
- Hosts Arc-enabled Kubernetes instance and the Kubernetes VM extension
- Must NOT be deleted unless decommissioning the cluster

Core Component: Kubernetes VM Extension

- Implements VM lifecycle actions
- Handles:
  - Power operations
  - Disk attach/detach
  - NIC attach/detach
  - GPU operations
- Executes ARM-triggered operations on the local cluster

Core Component: Custom Location

- Represents your Azure Local instance in Azure
- Used as the deployment target for VMs, NICs, disks, images
- One-to-one mapping with the resource bridge namespace

Core Component: Infrastructure Logical Network

- Created automatically during deployment
- Used by the Arc resource bridge
- Not user-modifiable

Azure Local VM Management Workflow

1. Deploy Azure Local → resource bridge + custom location created
2. Assign RBAC roles
   - Azure Stack HCI Administrator
   - Azure Stack HCI VM Contributor
   - Azure Stack HCI VM Reader
3. Create supporting VM resources:
   - Storage paths
   - VM images
   - Logical networks
   - Network interfaces
4. Create a VM
5. Use Azure Portal/CLI for all supported operations

Creating Storage Paths

- Storage paths store:
  - VM configuration files
  - OS & data disks
  - Image files
- Must reside on Cluster Shared Volumes (CSV)
- Should not use Infrastructure volumes

VM Images: Supported Sources

You can create VM images from:

- Azure Marketplace
- Azure Compute Gallery
- Azure Storage Account
- Local share (on-prem)
- Existing Azure Local VM’s OS disk

VM Image Requirements

- Only English (en-us) VHDs supported
- Must be generalized
- For VHDX:
  - Must be Gen 2
  - Must have Secure Boot enabled

Network Interfaces Overview

- Must be associated with a logical network
- NIC can be:
  - Static (automatic from an IP pool or manual assignment)
  - DHCP
- When created during VM deployment, NICs inherit configuration from the logical network

VM Creation Workflow

1. Click + Create VM
2. Enter:
   - VM name
   - Custom location
   - Security type
   - Storage path
3. Select image
4. Configure:
   - vCPU
   - Memory
   - Extensions
   - Networking
5. Configure proxy (optional)
6. Add disks (optional)
   - Disk name
   - Capacity
   - Provisioning type (static/dynamic)
   - Storage path
7. Add NICs (optional)
8. Review + create

Proxy for Azure Local VMs

- Public endpoint
- Proxy server
  - HTTP proxy
  - HTTPS proxy
  - No-proxy exclusions
  - Certificate file for trust establishment

VM Extensions (Optional)

Arc-enabled VMs support:

- Guest management
- Azure Monitor
- Defender for Cloud
- Custom extensions

Domain Join (Windows Only)

- Supports Active Directory domain join
- Requires:
  - UPN of domain user
  - Password
  - Domain or OU specification

Supported Cloud Initiated Azure Local VMs Operations

- Performed from the Azure Portal or Azure CLI:
  - Create
  - Resize CPU/memory
  - Add/remove disks
  - Add/remove NICs
  - Attach/detach GPU
  - Start/stop/restart/pause
  - Enable guest management
  - Install extensions
  - Connect via SSH/RDP
- Performing these operations via local tools risks state conflicts

Supported Local-Only Azure Local VMs Operations

- Performed via Hyper-V Manager/Failover Cluster Manager/PowerShell:
  - Rename VM
  - MAC address changes
  - NUMA config
  - Boot order changes
  - Secure boot enable
  - Checkpoints (HCI 2504+)
  - SCSI controller modifications
- These changes do not sync to Azure

Unsupported Operations

- Avoid entirely:
  - Changing VLAN ID
  - Storage live migration
  - Cloning or copying VM
  - Changing disk type
  - Cross-cluster live migration
- These operations break VM management state in Azure and can render VMs unmanageable.

Managed Identities

- Arc-enabled VMs include a system-assigned managed identity
- Allow access to:
  - Azure Key Vault
  - Azure Storage
  - Other identity-protected endpoints

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